

FINCORE ERP

Corporate Finance, Procurement & Accounting Management System

Version: 1.0

Project Type: Training / Learning ERP Project

Technology Stack: ASP.NET Core 8 Web API, Angular 18, MS SQL Server 2022

1. PROJECT OVERVIEW

Project Name

FINCORE ERP

Project Description

FINCORE ERP is a comprehensive Corporate Finance, Procurement, Budgeting, Revenue, Expense, and Accounting Management System designed to manage the complete financial lifecycle of an organization.

The system simulates enterprise-level ERP applications used by organizations to manage budgets, capital expenditures, operational expenditures, procurement activities, vendor management, payments, accounting, reporting, and financial controls.

The objective of the project is to gain hands-on experience with enterprise software development using modern technologies and industry-standard business processes.

2. PROJECT OBJECTIVES

- Manage annual and monthly budgets.
 - Track CAPEX (Capital Expenditure).
 - Track OPEX (Operational Expenditure).
 - Manage procurement lifecycle.
 - Generate Purchase Requisitions (PR).
 - Generate Purchase Orders (PO).
 - Generate Work Orders (WO).
 - Manage vendors and suppliers.
 - Record revenues and expenses.
 - Manage Accounts Payable.
 - Manage Accounts Receivable.
 - Maintain General Ledger.
 - Generate financial statements.
 - Support approval workflows.
 - Maintain audit logs.
 - Provide management dashboards and reports.
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3. SYSTEM USERS

Administrator

Responsible for system setup, security, and configuration.

Finance Manager

Responsible for budgeting, accounting, payments, and reporting.

Procurement Manager

Responsible for procurement operations and vendor management.

Department Head

Responsible for departmental approvals and budget control.

Employee

Responsible for raising requests and expense claims.

Auditor

Responsible for reviewing transactions and audit logs.

CFO

Responsible for high-value approvals and financial oversight.

4. TECHNOLOGY STACK

Frontend

- Angular 18+
- TypeScript
- Angular Material
- RxJS
- Bootstrap 5

Backend

- ASP.NET Core 8 Web API
- Entity Framework Core
- JWT Authentication
- Swagger / OpenAPI
- AutoMapper
- Fluent Validation

Database

- Microsoft SQL Server 2022

Reporting

- SSRS (SQL Server Reporting Services)
- PDF Report Generation
- Excel Export
- CSV Export

Authentication & Authorization

- JWT Authentication
- Role-Based Access Control (RBAC)

File Storage

- Local File Storage
- Document Upload Management

Architecture

- Clean Architecture
 - Repository Pattern
 - Unit of Work Pattern
 - Service Layer Pattern
 - Dependency Injection
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5. HIGH LEVEL MODULES

1. Authentication & User Management
 2. Organization Management
 3. Budget Management
 4. CAPEX Management
 5. OPEX Management
 6. Vendor Management
 7. Purchase Requisition Management
 8. RFQ Management
 9. Purchase Order Management
 10. Work Order Management
 11. Goods Receipt Note (GRN)
 12. Inventory & Asset Management
 13. Revenue Management
 14. Expense Management
 15. Accounts Payable
 16. Accounts Receivable
 17. Payment Management
 18. General Ledger
 19. Dashboard & Analytics
 20. Reporting
 21. Workflow Engine
 22. Audit Logs
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6. MODULE DETAILS

6.1 Authentication & User Management

Purpose

Manage users, roles, permissions, and authentication.

Features

- User Registration
 - Login
 - Logout
 - Password Reset
 - Role Assignment
 - Permission Assignment
 - User Status Management
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6.2 Organization Management

Purpose

Manage organization hierarchy.

Features

- Company Management
 - Branch Management
 - Department Management
 - Cost Centers
 - Profit Centers
-

6.3 Budget Management

Purpose

Manage yearly and monthly budgets.

Features

- Budget Creation
 - Budget Approval
 - Budget Revision
 - Budget Allocation
 - Budget Utilization Tracking
 - Budget Variance Analysis
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6.4 CAPEX Management

Purpose

Manage Capital Expenditure Requests.

Examples

- Machinery Purchase
- Vehicle Purchase
- Data Center Equipment
- Office Infrastructure

Features

- CAPEX Request Creation
 - Budget Validation
 - Approval Workflow
 - CAPEX Tracking
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6.5 OPEX Management

Purpose

Manage Operational Expenses.

Examples

- Rent
- Internet
- Utility Bills
- Software Licenses

Features

- Expense Recording
 - Expense Approval
 - Expense Monitoring
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6.6 Vendor Management

Purpose

Manage vendors and suppliers.

Features

- Vendor Registration
 - Vendor Classification
 - Vendor Document Upload
 - Vendor Performance Tracking
 - Vendor Evaluation
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6.7 Purchase Requisition Management

Purpose

Manage internal purchase requests.

Features

- Create PR
 - Submit PR
 - Approve PR
 - Reject PR
 - Track PR Status
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6.8 RFQ Management

Purpose

Manage vendor quotation process.

Features

- Create RFQ
 - Send RFQ
 - Receive Quotations
 - Compare Quotations
 - Select Vendor
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6.9 Purchase Order Management

Purpose

Generate and manage Purchase Orders.

Features

- Create PO
 - Approve PO
 - Cancel PO
 - Close PO
 - Generate PDF
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6.10 Work Order Management

Purpose

Manage service contracts.

Examples

- AMC
- Consultancy Services
- Maintenance Contracts

Features

- Create WO
 - Approve WO
 - Complete WO
 - Close WO
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6.11 Goods Receipt Note (GRN)

Purpose

Record receipt of purchased goods.

Features

- Receive Goods
 - Quantity Verification
 - Quality Verification
 - GRN Generation
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6.12 Inventory & Asset Management

Purpose

Manage organizational assets.

Features

- Asset Registration
 - Asset Tracking
 - Depreciation Calculation
 - Asset Disposal
-

6.13 Revenue Management

Purpose

Manage company income.

Features

- Sales Revenue Entry
- Service Revenue Entry
- Revenue Reporting
- Revenue Dashboard

6.14 Expense Management

Purpose

Manage employee and operational expenses.

Features

- Expense Claims
 - Expense Approval
 - Expense Reports
-

6.15 Accounts Payable

Purpose

Manage vendor liabilities.

Features

- Vendor Invoices
 - Payment Scheduling
 - Outstanding Payables
 - Aging Analysis
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6.16 Accounts Receivable

Purpose

Manage customer receivables.

Features

- Customer Invoices
 - Payment Collection
 - Outstanding Receivables
 - Aging Analysis
-

6.17 Payment Management

Purpose

Manage incoming and outgoing payments.

Features

- Payment Creation
 - Payment Approval
 - Reconciliation
 - Payment History
-

6.18 General Ledger

Purpose

Maintain accounting records.

Features

- Journal Entries
- Ledger Accounts
- Trial Balance
- Accounting Reports

6.19 Dashboard & Analytics

Purpose

Provide management insights.

KPIs

- Total Revenue
- Total Expenses
- Budget Utilization
- CAPEX Spend
- OPEX Spend
- Outstanding Payments
- Vendor Spend

6.20 Reporting

Reports

- Profit & Loss Statement
 - Balance Sheet
 - Cash Flow Statement
 - Budget Variance Report
 - CAPEX Report
 - OPEX Report
 - Vendor Spend Report
 - Revenue Report
 - Expense Report
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6.21 Workflow Engine

Purpose

Manage approvals.

Features

- Multi-Level Approval
 - Approval Escalation
 - Approval History
 - Workflow Configuration
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6.22 Audit Logs

Purpose

Track system activity.

Features

- User Activity Logs
 - Entity Change History
 - Approval History
 - Login Tracking
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7. PROCUREMENT FLOW

Purchase Requisition (PR)



Approval



RFQ Creation



Vendor Quotations



Vendor Comparison



Vendor Selection



Purchase Order (PO)



Goods Receipt Note (GRN)



Vendor Invoice



Payment



General Ledger Posting

8. CAPEX FLOW

CAPEX Request



Budget Validation



Approval



Procurement



Purchase Order



Asset Registration



Depreciation Tracking

9. OPEX FLOW

Expense Request



Approval



Expense Booking



Payment Processing



General Ledger Posting

10. API DESIGN

Base URL

/api/v1

MODULE 1 - AUTHENTICATION & USERS

POST /auth/login

POST /auth/refresh-token

POST /auth/change-password

POST /users

PUT /users/{id}

GET /users

GET /users/{id}

DELETE /users/{id}

POST /users/assign-role

MODULE 2 - ORGANIZATION

POST /companies

GET /companies

POST /departments

GET /departments

POST /cost-centers

GET /cost-centers

POST /profit-centers

GET /profit-centers

MODULE 3 - BUDGET

POST /budgets

PUT /budgets/{id}

GET /budgets

GET /budgets/{id}

POST /budgets/{id}/approve

GET /budgets/variance-report

GET /budgets/utilization

MODULE 4 - CAPEX

POST /capex

PUT /capex/{id}

GET /capex

GET /capex/{id}

POST /capex/{id}/submit

POST /capex/{id}/approve

POST /capex/{id}/reject

MODULE 5 - OPEX

POST /opex

PUT /opex/{id}

GET /opex

GET /opex/{id}

POST /opex/{id}/approve

POST /opex/{id}/reject

GET /opex/summary

MODULE 6 - VENDORS

POST /vendors

PUT /vendors/{id}

GET /vendors

GET /vendors/{id}

DELETE /vendors/{id}

POST /vendors/{id}/documents

GET /vendors/{id}/performance

MODULE 7 - PURCHASE REQUISITION

POST /purchase-requisitions

PUT /purchase-requisitions/{id}

GET /purchase-requisitions

GET /purchase-requisitions/{id}

POST /purchase-requisitions/{id}/submit

POST /purchase-requisitions/{id}/approve

POST /purchase-requisitions/{id}/reject

MODULE 8 - RFQ

POST /rfq

GET /rfq

GET /rfq/{id}

POST /rfq/{id}/send

GET /rfq/{id}/quotations

GET /rfq/{id}/comparison

POST /rfq/{id}/select-vendor

MODULE 9 - PURCHASE ORDER

POST /purchase-orders

PUT /purchase-orders/{id}

GET /purchase-orders

GET /purchase-orders/{id}

POST /purchase-orders/{id}/approve

POST /purchase-orders/{id}/cancel

POST /purchase-orders/{id}/close

GET /purchase-orders/{id}/pdf

MODULE 10 - WORK ORDER

POST /work-orders

PUT /work-orders/{id}

GET /work-orders

GET /work-orders/{id}

POST /work-orders/{id}/approve

POST /work-orders/{id}/complete

POST /work-orders/{id}/close

MODULE 11 - GRN

POST /grn

GET /grn

GET /grn/{id}

POST /grn/receive

MODULE 12 - ASSETS

POST /assets

PUT /assets/{id}

GET /assets

GET /assets/{id}

GET /assets/depreciation

POST /assets/{id}/dispose

MODULE 13 - REVENUE

POST /sales-invoices

GET /sales-invoices

GET /sales-invoices/{id}

GET /revenue/monthly

GET /revenue/summary

MODULE 14 - EXPENSES

POST /expense-claims

GET /expense-claims

GET /expense-claims/{id}

POST /expense-claims/{id}/approve

POST /expense-claims/{id}/reject

MODULE 15 - ACCOUNTS PAYABLE

POST /ap/invoices

GET /ap/invoices

POST /ap/invoices/{id}/approve

POST /ap/payments

GET /ap/outstanding

GET /ap/aging

MODULE 16 - ACCOUNTS RECEIVABLE

POST /ar/invoices

GET /ar/invoices

POST /ar/payments

GET /ar/outstanding

GET /ar/aging

MODULE 17 - PAYMENTS

POST /payments

GET /payments

POST /payments/{id}/approve

POST /payments/reconcile

MODULE 18 - GENERAL LEDGER

POST /gl/journals

GET /gl/journals

GET /gl/accounts

GET /gl/trial-balance

MODULE 19 - DASHBOARD

GET /dashboard/executive

GET /dashboard/finance

GET /dashboard/procurement

GET /dashboard/budget

MODULE 20 - REPORTS

GET /reports/profit-loss

GET /reports/balance-sheet

GET /reports/cash-flow

GET /reports/budget-variance

GET /reports/capex

GET /reports/opex

GET /reports/vendor-spend

MODULE 21 - WORKFLOW

POST /workflow/submit

POST /workflow/approve

POST /workflow/reject

GET /workflow/history/{entityId}

MODULE 22 - AUDIT

GET /audit/logs

GET /audit/user/{userId}

GET /audit/entity/{entityId}

11. DATABASE MODULES

Master Tables

- Users
- Roles
- Permissions
- Companies
- Departments
- CostCenters
- ProfitCenters
- Vendors
- Customers
- Employees
- AccountMasters
- BudgetCategories

Transaction Tables

- Budgets
- BudgetLines
- CapexRequests
- OpexRequests
- PurchaseRequisitions
- RFQs
- Quotations
- PurchaseOrders
- PurchaseOrderItems
- WorkOrders
- GRNs
- Assets
- RevenueEntries
- ExpenseClaims
- APInvoices
- ARInvoices
- Payments
- JournalEntries

Workflow Tables

- WorkflowDefinitions
- WorkflowSteps
- WorkflowHistory
- ApprovalLogs

Audit Tables

- AuditLogs
 - UserActivityLogs
 - NotificationLogs
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12. SOLUTION STRUCTURE

FinCoreERP

- └─ FinCoreERP.API
 - └─ FinCoreERP.Application
 - └─ FinCoreERP.Domain
 - └─ FinCoreERP.Infrastructure
 - └─ FinCoreERP.Persistence
 - └─ FinCoreERP.Identity
 - └─ FinCoreERP.Reporting
 - └─ FinCoreERP.Tests
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13. ANGULAR PROJECT STRUCTURE

src

- |— core
- |— shared
- |— layouts
- |— modules
- | |— dashboard
- | |— budget
- | |— capex
- | |— opex
- | |— vendors
- | |— procurement
- | |— purchase-orders
- | |— work-orders
- | |— revenue
- | |— expenses
- | |— accounts-payable
- | |— accounts-receivable
- | |— reports
- | |— administration
- |— assets

14. PROJECT OUTCOME

FINCORE ERP will provide a complete end-to-end Corporate Finance, Procurement, Budgeting, Revenue, Expense, Accounting, Reporting, Workflow, and Audit Management solution built using ASP.NET Core, Angular, and SQL Server while following enterprise-grade ERP design principles.

The project will serve as a full-scale training ERP application demonstrating real-world business processes, scalable architecture, secure APIs, modern frontend development, and enterprise software engineering best practices.